

Auto-Graded Homework that Invites Students to Think Broadly

Mark Eichenlaub · Art of Problem Solving

The default ask: "find the number"

Homework that grades itself usually presents a scenario and asks for one quantity. Such problems build mastery and tenacity — but a steady diet of them trains a narrow, procedural picture of what doing physics *is*: pick the equation, plug in, solve.

Find the tension in the rope.

Find the time to reach the ground.

Find the final speed.

Find the distance traveled.

Find the ...

The grader isn't the limit — the question format is

An auto-grader can check any answer with a well-defined key — not just numbers.

Change *what you ask for*, and the same instant-feedback machinery invites the patterns of thought we prize in rich human-feedback settings: designing experiments, choosing models, weighing evidence, critiquing calculations, diagnosing errors.

If the answer is ___, the thinking is ___

a number ± tolerance	execute a plan
items sorted into bins	apply a criterion consistently
two columns connected	map structure onto structure
all-that-apply choices	judge each claim on its own
a click on the image	commit to a spatial prediction
a ranking	compare cases without computing

Every format above runs auto-graded in AoPS Online courses today. The six problems at right are real homework, shown with their grading formats.

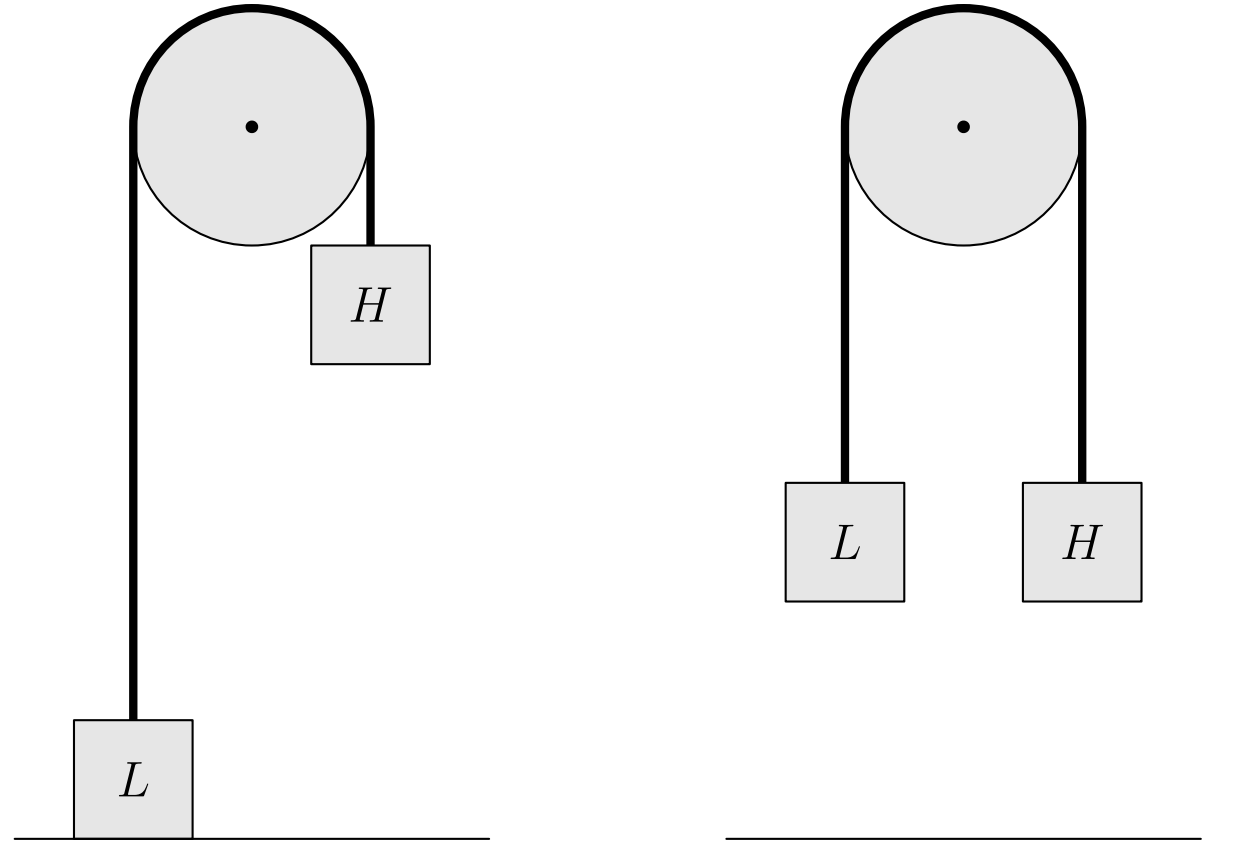
DESIGN AN EXPERIMENT

Physics 1: Mechanics · Acceleration

Make the Atwood machine measure g

You've modeled an Atwood machine's acceleration in terms of $r = m_L/m_H$. Now design an accurate experiment to measure g with it, using a stopwatch. For each pair of design choices, which gives a *more accurate* experiment?

- start the heavy box at the top — or halfway down?
- a single value of r — or many?
- one trial per r — or several?



start heavier box at top

start heavier box half way

answer = drag each choice into a bin

What students are doing: weighing longer timing windows, model-stressing ranges, and repeated trials — experimental-design judgment, graded with no paragraphs.

WEIGH EVIDENCE

Introduction to Physics · Thought Experiments

Which hypothesis does each observation support?

Galileo: all objects accelerate alike under gravity. Prof. Impetus (fictional): denser objects accelerate faster. Dr. Vacuum evaluates the evidence — but *never considers air drag*. Sort each observation by which hypothesis Dr. Vacuum will think it supports.

Galileo - Kepler's laws - hammer & feather, Moon - balloon falls on Moon	Impetus - crumpled paper falls faster - He balloon rises
Both lead balls land together	Neither skydiver, head-first

answer = drag observations into bins

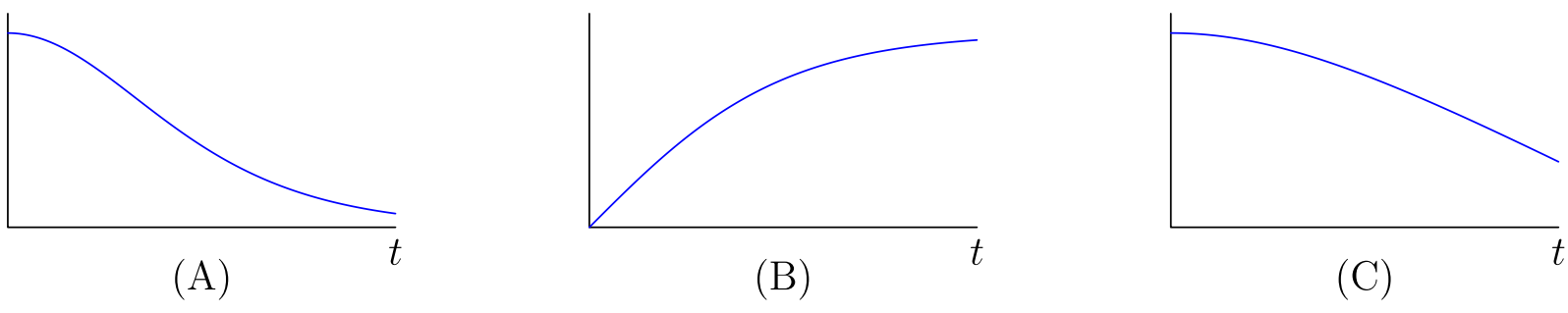
What students are doing: modeling a *flawed* reasoner's inferences — separating what an experiment seems to show from what it shows. Metacognition, graded as a partition.

INTERPRET GRAPHS

Physics 1: Mechanics · Forms of Energy

A peregrine falcon dives

A falcon dives from rest, approaching terminal velocity as air drag builds. Match each plot to the quantity it shows as a function of time: height, speed, or magnitude of acceleration.



answer = connect the columns

What students are doing: translating a physical story into qualitative features of curves — speed saturates, acceleration starts at g and dies away, height falls with steepening slope. No numbers come out, and only one matching survives.

DIAGNOSE A STUCK SOLVER

Introduction to Physics · Problem Solving

What should each student try next?

Five students are stuck on the classic string-around-the-Earth puzzle. Match each student's statement to their most productive next step.

"I don't understand this problem at all. I don't know where to start."	→	Restate it: what's unknown? what's given?
"My answer is 10^{12} m — bigger than Earth itself. Something seems wrong."	→	Check your methods and calculations
"I know it's about circumference, but I don't know what to do next."	→	Solve an easier related problem

answer = connect the columns

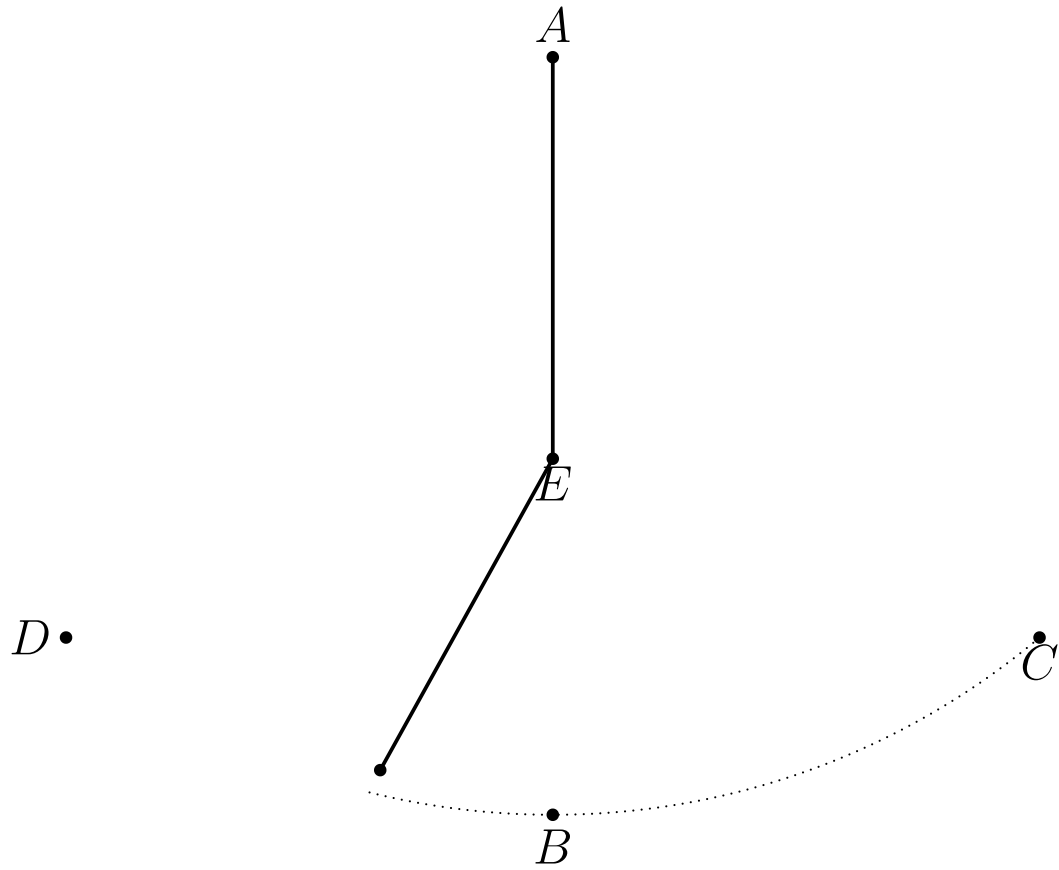
What students are doing: classifying *kinds of stuckness* and prescribing remedies — reasoning about the problem-solving process itself. Pólya, auto-graded.

COMMIT TO A PREDICTION

Introduction to Physics · Conservation Laws

Where does Galileo's pendulum stop?

A pendulum released from C swings through B , where the string catches on a peg at E and the bob swings on around the peg. Click the point in the image where the bob stops and turns around. (The point isn't necessarily labeled.)



answer = a click on the image

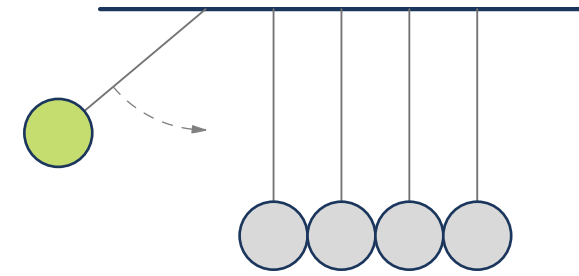
What students are doing: turning energy conservation into a *spatial* commitment — same release height, but on the tighter circle around the peg. Clicking D (same height, wrong circle) is a graded near-miss: right law, wrong constraint.

INTERPRET A CALCULATION

Physics 1: Mechanics · Collisions

The calculation disagrees with the cradle

For Newton's cradle, momentum + energy conservation alone predict something different from what the demo shows. Which interpretations are accurate?



- ☐ Momentum or energy must not actually be conserved in the collisions.
- ☒ Elastic collisions + momentum conservation are a good model — but not *enough*; more modeling assumptions are needed.
- ☒ The observed outcome is one solution of the conservation equations; there are many others.
- ☐ The cradle can't be modeled — only experiment can tell us what happens.

answer = select all that apply

What students are doing: deciding what a calculation does and doesn't show — built around a case where theory and demo *disagree*.

What are students doing?

Calculating is one row of this table — not the whole table:

- = apply known equations
- ✓ design a measurement
- ✓ interpret graphs & data
- ✓ build & choose models
- ✓ critique an argument
- ✓ weigh evidence
- ✓ interpret an equation
- ✓ diagnose an error
- ✓ estimate from experience
- ✓ reason about symmetry
- ✓ read & use notation
- ✓ reflect on learning

Every verb above appears in the 28 surveyed problems — each one auto-graded.

Design principles that emerged

- 1** Grade a decision, not a derivation. Bins, matches, and clicks capture judgments directly.
- 2** Put the misconception in the answer key. A wrong bin is a diagnosis ("lighter" at the valley bottom).
- 3** Ask about someone else's thinking. Flawed reasoners and stuck students invite metacognition.
- 4** Let the diagram be the answer. Click-on-image makes a spatial claim gradable.
- 5** Use cases where the model fails. Interpreting a calculation is a gradable skill.
- 6** Structure prevents lucky guessing. Forced pairs and one-to-one matches discipline the format.

What still needs humans

Proofs, lab write-ups, open design projects, and reflective writing need expert and peer feedback — nuance no key can hold. The claim here is narrower: the *graded* part of homework can invite the same patterns of thought, so the procedural diet isn't the default.

Take it with you



This poster + materials
markeichenlaub.github.io/aapt-sm26-poster

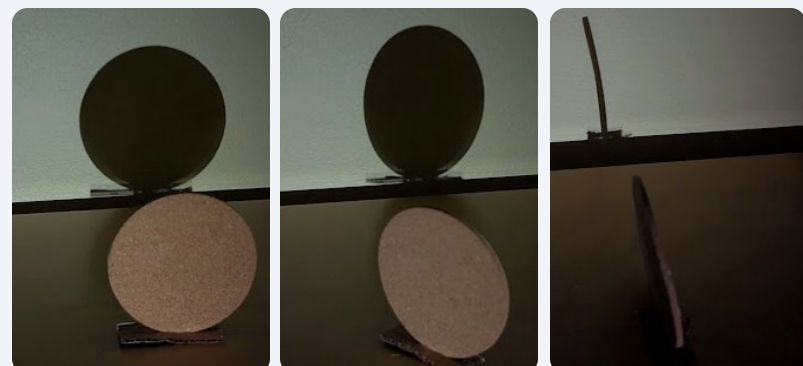


Try these problems, live
four auto-graded demos in your browser

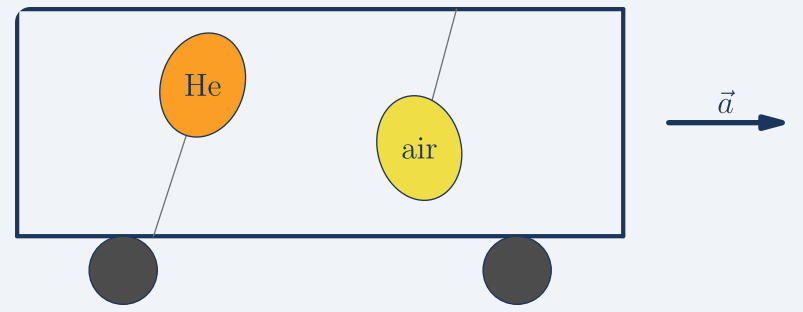


AoPS Online
artofproblemsolving.com/online

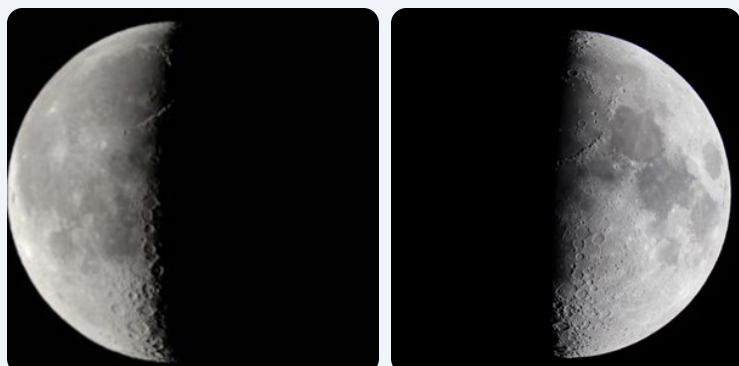
...and more, from a 28-problem survey across four AoPS physics courses



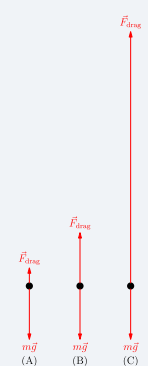
Reason with symmetry. Spin or roll a sphere — which actions change its shadow? Mostly: none. (Middle School Physics)



Predict the counterintuitive. Accelerate the car: the helium balloon leans *forward*. (Mechanics · Fluid Statics)



Reason in 3-D. Which quarter moon rises at sunset, and which at noon? Pure spatial reasoning, graded as a match. (Intro to Physics)



Read a diagram quantitatively. Which free-body diagram is terminal velocity — and which is the chute just opening? Arrow *lengths* carry the physics. (Mechanics)

$$\bar{q} \quad \Delta q \quad q = a \pm b$$

⇕

meanings

Read notation as language. Match decorations on q to what they say. Implicit literacy, made explicit. (Intro to Physics)

$$T = 2\pi\sqrt{\ell/g}$$

plot T^2 vs. ℓ

$$\Rightarrow \text{slope} = 4\pi^2/g$$

Linearize real data. Choose axes that straighten pendulum data, then read g off the slope — the working scientist's workflow. (Intro to Physics)